

# Murtaza Bootwala

Aspiring Computer Engineer and Robotics Enthusiast

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Aspiring computer engineer with hands-on experience in robotics fabrication, full-stack development, and effective instruction, blending technical proficiency with a passion for team development and problem-solving within dynamic environments.

## PROFESSIONAL EXPERIENCE

### Aero and Chassis Team | Cal Poly Racing

Sep 2025 – Present

San Luis Obispo, CA

- Assisted Aero & Chassis subgroups in manufacturing components for the Formula SAE vehicle, contributing to a 10–15% reduction in composite fabrication time through improved material handling
- Performed carbon layups, vacuum bagging, material preparation, and post-processing with <2% rework rate, ensuring quality and consistency in composite parts
- Assisted in prepping and assembling the carbon chassis tub with a focus on quality, accuracy, and structural performance
- Gained experience in composite fabrication workflows, safety procedures, and collaboration in a multidisciplinary engineering environment

### Swim Instructor | YMCA of Seattle

Jun 2023 – Sep 2025

Sammamish, WA

- Delivered tailored swim lessons to students across age groups and skill levels, adapting instructional techniques for effective knowledge transfer
- Monitored class safety, ensured adherence to established safety guidelines, and intervened promptly in emergency situations
- Cultivated an engaging and supportive environment that encourages student growth and builds water confidence
- Assessed individual progress and provided constructive feedback to help learners refine technique
- Coordinated scheduling and lesson plans with colleagues to maintain consistently high instructional standards

## PROJECTS

### MeetingTax | Next.js, AWS DynamoDB, Claude API, Vercel

- Built and deployed a full-stack B2B SaaS app for the H0: Hack the Zero Stack hackathon (Amazon & Vercel, \$80K prize pool) that calculates the real-time dollar cost of meetings based on attendee salaries, using DynamoDB for live data and Claude Haiku for AI-generated cost insights

### Sakkas Poster | Next.js, Supabase, Shopify Admin API, Meta Graph API, Claude API, Vercel

- Built and deployed a full-stack app that automates social media posting for a Shopify fashion retailer — auto-fetches product photos, generates on-brand captions with Claude, and posts or schedules carousel posts to Facebook and Instagram via a custom Supabase-based scheduler

### FoodSpect App | Flutter, Dart, Open Food Facts API

- Built a mobile app with a UPC barcode scanner that helps users with food allergies make safe dietary choices through real-time ingredient matching against a personalized allergy profile; placed 2nd at the WAForge Hackathon

## SKILLS

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| <ul style="list-style-type: none"><li>Composite Fabrication (Carbon Fiber, Vacuum Bagging)</li><li>CNC Machine Operation</li><li>Design for Manufacturing</li><li>Manufacturing Process Optimization</li><li>Java (JavaFX)</li><li>HTML/CSS &amp; JavaScript</li><li>Cloud Computing (AWS, Azure, Supabase)</li><li>Flutter &amp; Dart</li><li>Troubleshooting &amp; Problem Solving</li><li>Technical Instruction &amp; Communication</li><li>Failure Analysis</li></ul> | <ul style="list-style-type: none"><li>Mechanical Fabrication &amp; Assembly</li><li>CAD Modeling &amp; Technical Drawings</li><li>Robotics Fabrication &amp; Prototyping</li><li>Manufacturing Safety &amp; Compliance</li><li>Python</li><li>Next.js &amp; React</li><li>REST &amp; Third-Party API Integration</li><li>AI/LLM API Integration (Claude)</li><li>Team Leadership &amp; Mentorship</li><li>Hands-On Prototype Testing &amp; Validation</li></ul> |
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## EDUCATION

### California Polytechnic State University

San Luis Obispo, CA

Bachelor's degree, Computer Engineering • Expected Graduation: June 2029

## CERTIFICATES

- AWS Certified Cloud Practitioner | Amazon Web Services Training and Certification — August 15, 2025
- Harvard AI Bootcamp | Harvard University — December 2023
- Networking Basics | Cisco — March 27, 2025